The empirical recurrence for OEIS A229387, proved by transfer matrix

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Abstract

OEIS A229387 counts the distinct arrays obtained by applying a fixed local operation to every cell of an array over a bounded alphabet, and carries an empirical recurrence of order 6 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. What is counted is the IMAGE of a map, not its domain, so it is not a walk count in the obvious graph; determinising the machine that emits the derived array row by row turns it into one, on $S = 52$ states, after which the conjectured recurrence is settled by a finite exact computation.

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1 The conjecture

OEIS A229387 is "Number of n X 3 0..3 arrays of the sum of the corresponding element, the element to the east and the element to the south in a larger (n+1) X 4 0..1 array.". It has offset 1 and begins

$$48, 1064, 19124, 319340, 5212236, 84210828, 1353901580, 21715025932, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 25*a(n-1) -152*a(n-2) +144*a(n-3) -368*a(n-4) +1888*a(n-5) -1536*a(n-6) for n>9.

As of the "Last modified" line on the live entry (May 12 2014 13:17:10, revision 8) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

Let R be the set of rows of length 4 over $\{0, \ldots, 1\}$. For a larger array $A = (r_1, \ldots, r_{n+1})$ of $n + 1$ such rows let $\Phi(A)$ be the $n \times 3$ array with

$$\Phi(A)(i, j) \ = \ \mathrm{sum}\big(A(i, j),\, A(i, j + 1),\, A(i + 1, j)\big),$$

the element, the one to its east and the one to its south. The entry counts

$$a(n) \ = \ \big|\{\Phi(A)\}\big|,$$

the size of an IMAGE. That is the whole difficulty: two different larger arrays with the same derived array must be counted once, and a walk count on the larger arrays would count them twice.

Row i of $\Phi(A)$ reads rows i and $i+1$ of A and nothing else, so writing $\beta(r,x)$ for the derived row produced by two consecutive rows,

$$\Phi(r_1,\dots,r_{n+1}) = \big(\beta(r_1,r_2),\ \dots,\ \beta(r_n,r_{n+1})\big),$$

and there is no row emitted with nothing below it.

3 The image is a walk count after determinisation

For a set D of rows and a derived row b put

$$\delta(D,b) = \{x \in R :\ r \in D,\ \beta(r,x) = b\},$$

and let $D_0 = R$.

Lemma 1. *For every $t \geq 0$, $D_t = \delta(\dots\delta(D_0,b_1)\dots,b_t)$ is exactly the set of rows r_{t+1} completing a prefix $r_1,\dots,r_{t+1}$ whose derived rows are $b_1,\dots,b_t$. Consequently $a(n)$ is the number of sequences $(b_1,\dots,b_n)$ with $D_n \neq \emptyset$.*

Proof. Induction on t: $D_0 = R$ is the set of possible first rows, and $\delta(D_t,b)$ collects the rows r_{t+2} for which some consistent prefix ends in a row r_{t+1} with $\beta(r_{t+1},r_{t+2}) = b$. A derived array of n rows is realisable exactly when the corresponding D_n is nonempty, and distinct sequences are distinct derived arrays. □

So with M the adjacency matrix of the digraph whose vertices are the reachable nonempty sets and which carries one edge $D \to \delta(D,b)$ for each b with $\delta(D,b) \neq \emptyset$ — several b may lead to the same set, and then the edge is repeated, which is exactly right, since the derived arrays differ —

$$a(n) \;=\; \iota^\top M^n \tau, \qquad \iota = e_{D_0}, \quad \tau = (1,\dots,1)^\top.$$

There are $S = 52$ reachable sets, generated from D_0 outward, and a is C-finite.

4 The proof

Lemma 2. *Let $q(t) = t^6 - \sum_i c_i t^{6-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+6} - \sum_i c_i M^{j+6-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. □

Theorem 3.

$$a(n) \;=\; 25\,a(n-1) - 152\,a(n-2) + 144\,a(n-3) - 368\,a(n-4) + 1888\,a(n-5) - 1536\,a(n-6)$$

for every $n > 9$.

Proof. The vector $w = q(M)\tau$ was formed by 6 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots,$ again exactly; those integers vanish from the index corresponding to $n = 9+1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. The bound is exact: at $n = 9$ the recurrence fails on the entry's own published terms, so no larger range holds. □

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the machine is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 2 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A229387.

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[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.